MIL-L-45942A(WC)
7 September 1976
SUPERSEDING
MIL-L-45942(WC)
6 November 1972

MILITARY SPECIFICATION

LINK, CARTRIDGE: METALLIC BELT, 30MM, XM23

This specification is approved for use by Rock Island Arsenal, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of metallic belt link intended to hold 30mm ammunition in the form of flexible belts for use in the XM140, 30MM automatic gun.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military		
MIL-P-116	-	Preservation-Packaging, Methods of
MIL-W-13855	-	Weapons: Small Arms and Aircraft Armament Subsystem, General Specification for
MIL-I-45607	-	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Federal		
Fed. Std. No. 356	-	Commercial Packaging of Supplies and Equipment
Military		
MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1167	-	Ammunition Data Cards
MIL-STD-1168	-	Ammunition Lot Numbering

FSC 1305

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DRAWINGS

Rock Island Arsenal
F8447607 - Link, 30MM, XM23.

PUBLICATIONS

Rock Island Arsenal
P8447607 - Packaging Data Sheet for Link, Cartridge, Metallic Belt; 30MM, XM23

QE-G-DL
8447607 - Quality Engineering Data List, Link, 30MM, XM23

(Acceptance Inspection Drawings referenced in this specification form a part of QE-G-DL 8447607.)

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM)
ASTM E18 - Standard Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.1). Unless otherwise specified (see 6.1), the first article shall include a pilot pack (see 5.1).

3.2 Material, construction and design. The links shall conform to the material, construction and design requirements specified herein, on Drawing F8447607 and in MIL-W-13855.

3.2.1 Stripping force. The force required to strip Government standard 30mm cartridges from the links shall be not less than 25 pounds nor more than 45 pounds. Testing shall be as specified in 4.5.3.5.

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3.2.2 Permanent set. Links shall not take a permanent set of more than 0.005 inch when subjected to a tensile load of 100 pounds. Testing shall be as specified in 4.5.3.6.

3.2.3 Free hinging. When assembled into 100 round belts with Government standard 30mm dummy ammunition, the links shall hinge and fold over without binding, kinking or any interference. Testing shall be as specified in 4.5.3.7.

3.2.4 Twist. Links assembled into 100 round belts with Government standard 30mm cartridges shall twist through a 180° progressive turn of the linked cartridges with no evidence of partial or complete belt separation. Testing shall be as specified in 4.5.3.8.

3.2.5 Functioning. Links shall be capable of being assembled into belts of 100 rounds of Government standard 30mm, XM39 training ammunition and fired in an XM140 automatic gun without cracking, breaking, separating or malfunctioning. Testing shall be as specified in 4.4.

* 3.3 Marking. Links shall be marked in accordance with MIL-W-13855 and the link model number as specified on Drawing F8447607.

* 3.3.1 Lot identification. Links shall be identified with the inspection lot number as specified in MIL-STD-1168.

* 3.3.2 Data cards. Data cards conforming to MIL-STD-1167 shall be prepared for each lot of links and shall include the information specified in the contract (see 6.1).

3.4 Workmanship. Workmanship shall be in accordance with MIL-W-13855.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

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4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. The first article (initial production unit(s)) shall be submitted for inspection in accordance with the contract (see 6.1). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein, the functioning test specified in 4.6.9 and such other inspection as necessary to determine that all requirements of the contract have been met.

* 4.4.1 Examination and test. Five components from the first article shall be inspected for all unlisted drawing dimensions and tests.

4.5 Quality conformance inspection.

4.5.1 Inspection lot. Unless otherwise specified herein, the formation, size and presentation of inspection lots shall be in accordance with MIL-STD-105 and MIL-W-13855. Samples shall be withdrawn in accordance with MIL-STD-105 from packaged links. Inspection lots shall not exceed 500,000 links per lot.

4.5.1.1 Heat treat lot. The heat treat lot shall be limited to one heat of steel and shall consist of links, heat treated in a batch type furnace at one charge or in a continuous type furnace during an uninterrupted period extending over not more than one eight hour shift.

4.5.2 Examination. Examination of links shall be performed as specified herein and in MIL-W-13855.

4.5.2.1 Classification of defects. Classification of defects and acceptable quality levels (AQLs) for examination in accordance with MIL-STD-105 shall be as specified herein and shall include the following provisions:

a. Unless otherwise specified, AQLs apply to each listed characteristic.
b. Where a Government drawing is listed as the method of inspection, the contractor shall use the inspection equipment fabricated in accordance with the applicable drawing. In the event the contractor desires to use alternate equipment which equals or exceeds the accuracy requirements of the specified drawing, it shall be submitted to the Government for approval.
c. Where Standard Measuring Equipment (SME) is specified as the method of inspection, the contractor may use any type of industry developed, commercially available, multi-usage equipment or special inspection equipment approved by the Government.

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* d. Where Visual is specified as the method of inspection of protective coating, the coating shall be visually examined for completeness, uniformity in appearance and color, for freedom from pits, corrosion, scratches and worn or bare spots.

* e. Where Visual is specified as the method of dimensional inspection, the sample shall be either scaled or compared with a specimen of known acceptable quality.

4.5.2.1.1 Link, 30MM, XM23, Drawing F8447607.

	Categories	Characteristic	AQL	Inspection Method
	Critical:	None defined.		
	Major:			
	101	Straightness of spine.	0.65	8440250
*	102	Location, tab.	0.65	8440247, 8440065 & 8440244
	103	Length of spine.	0.65	8440249
	104	Location, center loop.	0.65	8440249
	105	Location, rear loop.	0.65	8440249
	106	Location, front loop.	0.65	8440249
	107	Location, connecting loop	0.65	8440249
	108	Location, tang.	1.0	SME
	109	Height of tab.	1.0	SME
	110	Front loop diameter.	1.0	SME
	111	Rear loop diameter.	1.0	SME
	112	Width connector slot.	1.5	SME
	113	Width tang opening.	1.5	SME
	114	Stock thickness.	1.5	SME
*	115	Location, Arms	1.0	8440249
	Minor:			
	201	Height of tang.	2.5	SME
	202	Symmetry of tang opening.	2.5	SME
	203	Symmetry of spine.	2.5	SME
	204	Symmetry of center spine.	2.5	SME
	205	Symmetry of spine end.	2.5	SME
	206	Location, end of center loop.	2.5	SME
	207	Width, connector loop.	2.5	SME
	208	Improper radius, rear loop.	2.5	SME
	209	Improper radius, front loop.	2.5	SME
	210	Improper radius, connector loop.	2.5	SME
	211	Incorrect or missing protective coating.	2.5	Visual
	212	Incorrect or missing identification.	2.5	Visual
	213	Workmanship. (3.4)	4.0	Visual

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4.5.2.1.2 Packaging. Unless otherwise specified in each listed characteristic, the packaging requirements are specified on Packaging Data Sheet P8447607 (see 5.2). The inspection method shall be visual.

Categories	Characteristics	AQL
Major:		
101	Improper level of preservation and packing (see procurement documents).	1.0
102	Incorrect or illegible marking.	1.0
103	Contaminated links.	1.5
104	Improper containers.	1.5
105	Container interior contaminated with dirt, lint or other foreign matter.	1.5
106	Improperly oriented links in link cartons.	1.5
107	Improperly oriented link cartons in shipping container.	1.5
108	Incorrect quantity of links per shipping container.	1.5
109	Improper closure of shipping container.	1.5
110	Incorrect or missing inspection lot number on each carton of links.	1.5
* 111	Improper preservative application and drainage (Improper drainage is the presence or accumulation of liquid oil which is not held as an integral part of the phospate coating and is evident as droplets, puddles or as an oil film so wet as to be moving visually.)	1.5
Minor:		
201	Workmanship.	4.0

4.5.3 Testing.

4.5.3.1 Certification. Unless otherwise specified, the contractor shall furnish the Government representative with certified statements that each inspection lot conforms to the materials and processes specified on the applicable drawings and specification.

* 4.5.3.2 Accelerated corrosion testing. Sample links shall be selected from inspection lots for each processing bath prior to application of supplementary oil treatment. Lot formation and sampling shall be in accordance with MIL-P-16232 whether manual basket dip or continuous conveyor equipment is used. The sample links shall be tested for compliance with the requirements of MIL-P-16232 using the test methods specified in 4.6.1.

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* 4.5.3.3 Hardness testing. Sample links shall be taken from each heat treat lot in accordance with MIL-STD-105, using a single sampling plan, normal inspection, special inspection level S-4 and tested for hardness using the test method specified in 4.6.2. Acceptance shall be based on an AQL of 0.40 percent defective.

4.5.3.4 Decarburization testing. A sample of 5 links shall be selected at random from each heat treat lot and tested for decarburization using the test method specified in 4.6.3. Failure of any link to meet the requirements shall cause rejection of the heat treat lot.

* 4.5.3.5 Stripping force testing. A sample of 200 links shall be taken from each inspection lot and tested for stripping force (see 3.2.1) using the test method specified in 4.6.4. Failure of three or more links shall be cause for rejection of the inspection lot.

* 4.5.3.6 Permanent set testing. A sample of 32 links shall be taken from each heat treat lot and tested for permament set (see 3.2.2) using the test method specified in 4.6.5. Failure of any link shall be cause for rejection of the heat treat lot.

4.5.3.7 Free hinging testing. A separate sample of 800 links from each inspection lot shall be tested for free hinging (see 3.2.3) using the test method specified in 4.6.6. Failure of any link shall be cause for rejection of the lot.

4.5.3.8 Twist testing. The 800 link sample specified in 4.5.3.7 shall be tested for twist (see 3.2.4) using the test method specified in 4.6.7. Failure of any link shall be cause for rejection of the lot.

4.5.3.9 Determination of cleanliness testing. Links shall be tested for cleanliness using the sampling procedure and test method specified in 4.6.8.

4.5.4 Inspection equipment.

4.5.4.1 Acquisition, maintenance and disposition. Unless otherwise specified (see 6.1) responsibility for acquisition, maintenance and disposition of acceptance inspection and test equipment prescribed on the Quality Engineering Data List QE-G-DL8447607 and for all other inspection equipment required to perform inspection prescribed by applicable specifications shall be in accordance with MIL-I-45607.

4.5.4.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it must be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.6 Test methods.

* 4.6.1 Accelerated corrosion test. Sample links shall be tested for corrosion resistance of the applied protective coating using either salt spray (fog) test or Sprinfield Immersion Test specified in MIL-P-16232.

* 4.6.2 Hardness test. The sample links shall be tested for hardness requirements using ASTM E18.

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* 4.6.3 Decarburization test. A cross section of the sample links shall be polished, etched with a 2 to 5 percent nital solution and examined under a microscope of not less than 500 power to determine compliance with decarburization requirements of Drawing F8447607.

4.6.4 Stripping force test. Sample links shall be tested using the inspection equipment and test method specified in QE-G-DL8447607.

* 4.6.5 Permanent set test. Sample links shall be marked for identification and the 2.00 inch link pitch dimension measured and recorded. A 100 pound tensile load shall be applied to the link in the plane of the link pitch dimension. The link shall then be remeasured along the pitch dimension. Upon completion of each permanent set test the links shall be scrapped.

4.6.6 Free hinging test. Sample links shall be tested by assembling the links into 100 round belts using Government standard 30mm dummy cartridges. The belt shall then be laid out horizontally to it full length and one end drawn back and forth over the remainder of the belt.

4.6.7 Twist test. Sample links shall be tested by assembling the links with Government standard 30mm dummy cartridges into belts of 100 linked cartridges. The belt shall be laid out horizontally to its full length, with the open side of the links down, and one end of the belt flipped sharply by hand 180° to its other side (open side of the links up) so that the twisting action will progressively move along the belt to the free end.

4.6.8 Determination of cleanliness test. The links shall be tested for cleanliness requirements using MIL-P-116.

4.6.9 Functioning test. A sample of 600 links taken from the first article shall be tested by assembling the links into 6 belts of 100 rounds each using Government Standard 30mm, XM639 training ammunition. The belts shall be fired in the XM140 automatic gun in intermittent bursts of 5 to 15 rounds with 20 second intervals between bursts.

* 5. PACKAGING

5.1 Pilot pack. A pilot pack shall consist of links preserved in accordance with Packaging Data Sheet P8447607 for the level of protection specified in the contract (see 6.1), packed level B and forwarded as specified in the contract (see 3.1).

5.2 Levels A and B. Preservation, packing and marking shall be in accordance with Packaging Data Sheet P8447607 for the level of protection specified in the contract (see 6.1).

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5.3 Commercial packaging. Preservation, packing and marking shall be in accordance with Fed. Std. No. 356.

6. NOTES

6.1 Ordering data. Procurement documents should specify:

6.1.1 Procurement requirements:

a. Title, number and date of this specification.
b. List of drawings, specifications and publications pertinent to the link showing applicable revision dates.
c. Requirements for submission of first article and pilot pack (see 3.1, 4.4 and 5.1).
d. Data card information required (see 3.2.2).
e. Responsibility for acquisition, maintenance and disposition of inspection equipment, if other than as specified (see 4.5.4.1).
f. Responsibility for furnishing dummy cartridges.
g. Selection of applicable levels of protection (A or B), or commercial packaging (5.1, 5.2 and 5.3).

6.1.2 Contract data requirements. When warranted, requirements for reports of the results of examination and testing should be specified on a DD Form 1423 included in the contract.

6.2 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208 on the Management Control Systems Summary List, DD Form 1660.

6.3 Unless otherwise specified (see 6.1), the contract should specify the application of MIL-I-45607 on the Management Control Systems Summary List, DD Form 1660.

6.4 Supersession data. This specification includes the requirements of U.S. Army Weapons Command Purchase Description, USAWECOM PD-12 dated 2 August 1968.

6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - WC

Review activity:
Army - MU

Preparing activity:
Army - WC

Project Number:
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DOCUMENT IDENTIFIER AND TITLE

MIL-L-45942A(WC); Link, Cartridge: Metallic Belt, 30MM, XM23

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIE*

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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   ☐ YES ☐ NO *(If "Yes", in what way?)*

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